

Exact Rational Arithmetic for Sequential Computation

VDR Applied to Twenty Domains Where Decimal Truncation Compounds

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Abstract

Decimal arithmetic — including arbitrary-precision libraries like `mpmath` — represents numbers in base 10. The only fractions with terminating decimal representations are those whose denominators factor exclusively into 2s and 5s. Every other fraction — $1/3$, $1/7$, $1/11$, $1/13$, every fraction whose denominator contains any prime factor other than 2 or 5 — becomes an infinite repeating decimal that must be truncated. This truncation is not a precision issue solvable by carrying more digits. It is a structural incompatibility between the representation and the value.

This paper identifies twenty computational domains where this structural incompatibility produces materially wrong results, and demonstrates that VDR exact rational arithmetic [VDR-1] eliminates the problem entirely. VDR represents $1/3$ as $[1, 3, 0]$ — exact, with zero truncation. Chains of operations on exact rationals produce exact rationals. The denominator grows but nothing is discarded. This is not “more precision.” It is a categorically different relationship with the numbers.

No prior reading is required. VDR arithmetic is summarized where first used; full specifications are in [VDR-1] and [VDR-14].

1. The Decimal Trap

1.1 Why $10 = 2 \times 5$ Is the Problem

A fraction a/b has a terminating decimal representation if and only if every prime factor of b (after reducing a/b to lowest terms) is 2 or 5. This is because decimal notation is positional base 10, and $10 = 2 \times 5$. The denominator must divide some power of 10, which means it must be of the form $2^m \times 5^n$.

$1/2 = 0.5$. Terminates. $1/4 = 0.25$. Terminates. $1/5 = 0.2$. Terminates. $1/8 = 0.125$. Terminates. $1/20 = 0.05$. Terminates. These are the lucky fractions — the ones whose denominators happen to share factors with our counting base.

$1/3 = 0.333\dots$ Repeats forever. $1/7 = 0.142857142857\dots$ Repeats with period 6. $1/11 = 0.090909\dots$ Repeats with period 2. $1/13 = 0.076923076923\dots$ Repeats with period 6. $1/6 = 0.1666\dots$ Repeats. Even fractions with mixed factors — $1/6 = 1/(2 \times 3)$ — repeat because of the factor of 3.

Among the first 20 positive integers used as denominators, only 1, 2, 4, 5, 8, 10, 16, 20 produce terminating decimals. The other 12 — 3, 6, 7, 9, 11, 12, 13, 14, 15, 17, 18, 19 — produce infinite repeating decimals. Sixty percent of small denominators are incompatible with decimal representation.

1.2 What Arbitrary Precision Does Not Solve

Arbitrary-precision decimal libraries like `mpmath` address the problem by carrying hundreds or thousands of digits. This reduces the magnitude of the truncation error per operation but does not eliminate it.

Multiply two 1000-digit truncated values: the exact product needs 2000 digits. `mpmath` truncates back to 1000. Chain 50 operations: 50 truncations, each discarding information from the bottom of the intermediate result. The per-operation loss is small. The accumulated loss over a chain is unknown — `mpmath` does not track what it discarded. The user sees 1000 digits and assumes they are all correct. Some of them are not, and the user cannot determine which ones without repeating the computation at 2000 digits, which defers the same problem.

The fundamental issue is not how many digits are carried. It is that the representation forces truncation on values that are exactly representable in other forms. $1/3$ is three characters: 1, /, 3. In VDR it is three integers: [1, 3, 0]. In decimal it is an infinite sequence of digits that can never be written in full. Carrying more digits is an arbitrarily expensive workaround for a representation deficiency.

1.3 What VDR Does Instead

VDR represents every rational number as three integers: Value, Denominator, Remainder [VDR-1]. $1/3$ is [1, 3, 0]. $1/7$ is [1, 7, 0]. $691/2730$ is [691, 2730, 0]. The denominator is stored as an integer, not decomposed into a decimal expansion. No truncation, no repeating digits, no lost information.

Arithmetic on VDR rationals is exact. $[1, 3, 0] \times [1, 7, 0] = [1, 21, 0]$. Chain 50 operations: exact. Chain 1000 operations: exact. The denominator grows — 21, 441, 9261, ... — but nothing is discarded. Every digit of every intermediate result is preserved in the numerator and denominator integers.

When precision must be bounded — for GPU execution using Q335 fixed-frame arithmetic [VDR-14, VDR-18] — the Remainder slot carries exactly what was not absorbed by the denominator frame. The Remainder is not rounding error. It is first-class structural information: visible, queryable, exact. The system knows precisely what it approximated and by how much.

2. Continued Fraction Arithmetic

2.1 The Problem

The Gauss continued fraction for ratios of hypergeometric functions produces convergents that are ratios of integers with no factors of 2 or 5. Computing ${}_2F_1(1/2, 1/2; 1; k^2)$ for the complete elliptic integral $K(k)$ at rational k produces convergents like 7/22, 103/323, 1507/4726. Every one of these is an infinite repeating decimal.

The continued fraction recurrence $p_n = a_n p_{n-1} + p_{n-2}$ is pure integer arithmetic. Each convergent p_n/q_n is an exact rational. But in decimal, each convergent is truncated, and the recurrence that produces the next convergent amplifies the truncation because it multiplies the previous convergent by the next partial quotient and adds the one before that — two operations on truncated values producing a more truncated result.

2.2 The VDR Solution

Each convergent is $[p_n, q_n, 0]$ where p_n and q_n are exact integers. The recurrence is exact integer arithmetic. After 200 convergents, every digit is correct. VDR-2 gym domain 3 verified: Gauss-Kuzmin distribution statistics show coefficient 1 appearing at approximately 41% (predicted 41.5%), which requires exact counting over exact convergents [VDR-2].

2.3 Where It Matters

The quality of a rational approximation p/q to an irrational α is measured by $|\alpha - p/q|$ — a subtraction of two nearly equal numbers. In decimal, this is catastrophic cancellation: the leading digits cancel and the remaining digits are contaminated by truncation. In VDR, the subtraction is exact: $(\alpha_n \cdot q - p) / (D \cdot q)$ where α_n is the Newton iterate at chosen depth, all exact. Number theory, Diophantine approximation, and irrationality measure computation all depend on this exact subtraction.

3. Bernoulli Numbers and Zeta Values

3.1 The Problem

Bernoulli numbers B_n grow factorially and have denominators governed by the von Staudt-Clausen theorem: the denominator of B_{2n} is the product of all primes p such that $(p-1)$ divides $2n$. $B_{12} = -691/2730$. The denominator $2730 = 2 \times 3 \times 5 \times 7 \times 13$. In decimal: $-691/2730 = -0.253113553113553113\dots$ repeating with period 12.

For B_{30} the denominator is $14322 = 2 \times 3 \times 2387$ and the numerator is 8615841276005 . The period of the decimal repeating block is determined by the prime 2387, potentially thousands of digits long. mpmath truncates this after N digits regardless of the period.

3.2 The VDR Solution

$B_{12} = [-691, 2730, 0]$. Exact. Multiply by any rational: exact. The zeta value relationship $\zeta(2n) = (-1)^{n+1} \cdot B_{2n} \cdot (2\pi)^{2n} / (2 \cdot (2n)!)$ is computed with exact rational Bernoulli numbers times Q335 powers of π [MATH-4]. The only approximation is the Q335 representation of π , which has a known, exact, 100-digit precision floor. The Bernoulli factor contributes zero error.

3.3 Where It Matters

Analytic number theory, computing special function values, and perturbative quantum field theory (Bernoulli numbers appear in Euler-Maclaurin summation of Feynman integrals) all depend on exact Bernoulli number arithmetic. The decimal trap makes high-index Bernoulli numbers progressively harder to work with as the denominators accumulate more odd prime factors. VDR eliminates this progression entirely.

4. Hilbert Matrix Computations

4.1 The Problem

The Hilbert matrix $H_{ij} = 1/(i+j-1)$ is the canonical ill-conditioned matrix. Every entry is a unit fraction: $1/1, 1/2, 1/3, 1/4$, and so on. In decimal, $1/3 = 0.333\dots$, $1/7 = 0.142857\dots$, $1/11 = 0.090909\dots$ — all infinite repeating decimals, all truncated.

The matrix is notoriously ill-conditioned. The condition number of H_n grows exponentially: approximately $e^{3.5n}$. This means that every digit of truncation error in the entries is amplified by the condition number in the inverse. mpmath at 50 digits computes H_5 inverse with residual approximately 10^{-40} — but there are digits wrong at the bottom. H_{10} residual is approximately 10^{-20} . H_{15} needs 100+ digits for a meaningful result. H_{20} at 200 digits produces a result where you cannot verify which digits are correct without going to 400 digits, and those require 800 to verify.

4.2 The VDR Solution

H_5 inverse residual = 0. Not approximately zero. Zero. $H_{10} = 0$. $H_{20} = 0$. $H_{30} = 0$. Every entry of H^{-1} is an exact integer — a known mathematical result: Hilbert inverse entries are always integers. VDR computes them as exact integers [VDR-1, VDR-2]. The determinant of H_4 is exactly 1/6048000. VDR returns [1, 6048000, 0]. The 40-operation matrix multiply roundtrip $H \times H^{-1} \times H \times \dots$ produces zero drift [VDR-1].

4.3 Where It Matters

Numerical analysis benchmarking, condition number studies, and any application where the matrix entries are exact rationals: stiffness matrices in structural engineering, Gram matrices in approximation theory, kernel matrices in machine learning with rational kernel parameters. In every case, the entries have denominators with prime factors other than 2 and 5, and decimal truncation interacts with ill-conditioning to produce unknowably wrong results.

5. Exact Eigenvalue Problems

5.1 The Problem

The eigenvalues of a 2×2 rational matrix $[[a,b],[c,d]]$ are $(a+d)/2 \pm \sqrt{((a-d)^2/4 + bc)}$. The discriminant is an exact rational. If it is a perfect square rational, the eigenvalues are exact rationals. If not, they involve the square root of a rational — representable as a VDR functional remainder with exact rational convergents at each Newton depth [VDR-1].

For the matrix $[[1/3, 1/7], [1/11, 1/13]]$, the discriminant is $(1/3 - 1/13)^2/4 + 1/(7 \cdot 11) = (10/39)^2/4 + 1/77 = 100/6084 + 1/77 = 13784/468468 = 3446/117117$. In decimal: 0.029422... repeating. mpmath takes $\sqrt{0.029422\dots}$ and truncates.

5.2 The VDR Solution

Discriminant = [3446, 117117, 0]. Exact. $\sqrt{\text{discriminant}}$ via Newton iteration: exact rational at each depth, converging quadratically. Eigenvalues are $(\text{trace}/2) \pm \sqrt{\text{discriminant}}$, each an exact rational expression. Eigenvector components are ratios of exact rationals. Orthogonality verified by exact dot product = 0, not dot product $\approx 10^{-15}$ [VDR-2].

VDR-2 gym domain 11 computed exact Markov chain steady state $[2/3, 1/3]$ — the eigenvector for eigenvalue 1. The steady state components have denominator 3, an infinite repeating decimal in base 10, exact in VDR.

5.3 Where It Matters

Quantum mechanics (2-level systems are 2×2 eigenvalue problems), stability analysis (eigenvalue sign determines stability — the critical case is the decimal trap at its worst, where the eigenvalue is near zero and truncation determines the sign), vibration analysis

(natural frequencies are eigenvalues of mass-stiffness systems), and Markov chain analysis.

6. Gaussian Elimination Pivot Ratios

6.1 The Problem

Gaussian elimination produces pivot elements that are ratios of determinants of submatrices. For rational input, every pivot is rational. But the denominators grow combinatorially. For an $n \times n$ matrix, pivot denominators can have $O(n)$ distinct prime factors, most of which are not 2 or 5.

Each elimination step divides by the pivot. If the pivot is $7/13$, dividing by it means multiplying by $13/7$. In decimal, $13/7 = 1.857142857142\dots$ repeating with period 6. This truncated value is multiplied by another truncated entry. The error compounds through $O(n^2)$ operations. For $n=20$, approximately 400 operations on truncated decimals produce a result with unknowably many wrong digits.

6.2 The VDR Solution

$13/7 = [13, 7, 0]$. Division by $[7, 13, 0]$ is exact rational arithmetic. 400 operations: exact. VDR-13 documented: H_{30} inverse via Gaussian elimination has pivot denominators up to approximately 185 digits [VDR-13]. mpmath at 200 digits would lose 15 digits of precision through the elimination. VDR loses zero.

6.3 Where It Matters

Every application of linear algebra on rational data. Structural engineering (stiffness matrices with rational entries), circuit analysis (impedances as rational functions of component values), least-squares fitting with rational data points, and exact rank computation. Float rank is always numerically full — a zero singular value appears as 10^{-15} and is indistinguishable from a small but nonzero value. VDR rank is exact: a zero entry is $[0, 1, 0]$ and a nonzero entry is not.

7. Probability Chains and Bayesian Updates

7.1 The Problem

Bayes' theorem with rational priors and likelihoods produces rational posteriors. Sequential Bayesian updating applies Bayes N times, each step's posterior becoming the next step's prior. The denominators grow because $P(B) = \sum P(B|A_i) \cdot P(A_i)$ involves sums of products of rationals, producing denominators with prime factors from every term.

$P(\text{disease}) = 1/10000$. $P(\text{positive}|\text{disease}) = 99/100$. $P(\text{positive}|\text{healthy}) = 1/100$.
 $P(\text{disease}|\text{positive}) = (99/100 \cdot 1/10000) / (99/100 \cdot 1/10000 + 1/100 \cdot 9999/10000)$. In

decimal, every intermediate value is truncated. After 10 updates with different evidence, the posterior has unknown digits at the bottom. After 100 updates, the accumulated truncation is substantial.

7.2 The VDR Solution

$P(\text{disease}) = [1, 10000, 0]$. $P(\text{positive}|\text{disease}) = [99, 100, 0]$. The posterior is an exact rational after normalization. After 10 updates: exact. After 100 updates: exact. The denominators are large but every digit is correct. VDR-2 gym domain 11 verified: binomial PMF with $n=10$, $p=1/3$ sums to exactly 1. Sequential Bayesian posterior computed as exact $6/7$ [VDR-2]. `mpmath` gives $0.857142857142\dots$ truncated.

7.3 Where It Matters

Medical diagnostics (sequential test results updating disease probability — getting this wrong has clinical consequences), spam filtering (sequential word observations updating spam probability), sensor fusion (sequential sensor readings updating state estimates), and forensic statistics (sequential evidence updating guilt probability — getting this wrong has legal consequences).

8. Markov Chain Steady States

8.1 The Problem

The steady state of a Markov chain is the eigenvector of the transition matrix for eigenvalue 1. For a k -state chain with rational transition probabilities, the steady state components are rational numbers whose denominators involve all the prime factors from the transition matrix entries.

Computing A^n for large n (power method) involves matrix multiplications where every entry is a truncated decimal. After $n=100$ multiplications, the last several digits of every entry are wrong. Alternatively, solving $(A-I)x = 0$ by Gaussian elimination on truncated entries produces the same problem described in Section 6.

For a 5-state chain with transition probabilities like $1/7$, $3/11$, $2/13$, the steady state components have denominators involving 7, 11, and 13 — all infinite repeating decimals in base 10.

8.2 The VDR Solution

Transition matrix entries are exact rationals. Matrix power A^n is exact. The steady state is the exact eigenvector. VDR-2 gym domain 11 computed exact steady state $[2/3, 1/3]$ for a 2-state chain [VDR-2]. VDR-3 gym domain 16 computed exact PageRank summing to exactly 1 [VDR-3].

Mixing time analysis depends on the second-largest eigenvalue magnitude $|\lambda_2|$. Computing $|\lambda_2|$ exactly (for small chains) or to arbitrary precision (via Newton functional remain-

der) means the mixing time bound is exact, not contaminated by arithmetic error in the eigenvalue computation.

8.3 Where It Matters

PageRank computation, queueing theory (steady-state queue lengths), population genetics (allele frequency evolution), and MCMC convergence diagnostics. In each case, the steady state is a set of exact rationals that decimal representation silently corrupts.

9. Elliptic Curve Arithmetic

9.1 The Problem

Points on an elliptic curve $y^2 = x^3 + ax + b$ over the rationals have coordinates that are rational numbers. Point addition involves $\lambda = (y_2 - y_1)/(x_2 - x_1)$, $x_3 = \lambda^2 - x_1 - x_2$, $y_3 = \lambda(x_1 - x_3) - y_1$. Every operation produces rationals. But the denominators grow rapidly: after n point doublings, coordinates can have denominators with $O(4^n)$ digits due to height growth.

After 20 doublings, coordinates have approximately 10^6 digit denominators. `mpmath` truncates these to N digits. The truncated point is not on the curve — $y^2 \neq x^3 + ax + b$ due to truncation. Group law properties cannot be verified on truncated points.

9.2 The VDR Solution

Each point addition is exact rational arithmetic. After 20 doublings, the coordinates have enormous denominators but are exactly on the curve: $y_3^2 = x_3^3 + ax_3 + b$ as exact VDR equality. Group law associativity $(P+Q)+R = P+(Q+R)$ is verified as exact structural equality, not approximate numerical agreement. The height of the point (logarithm of denominator) is exactly computable.

9.3 Where It Matters

Elliptic curve cryptography (point multiplication nG must be exact), Birch and Swinnerton-Dyer conjecture computations (rational point heights), integer factorization via Lenstra's ECM, and computing Mordell-Weil groups. In each case, the essential computation is an exact chain of rational arithmetic on points whose coordinates have denominators with many distinct prime factors.

10. Modular Arithmetic and Chinese Remainder Theorem

10.1 The Problem

CRT reconstructs an integer from its residues modulo several coprime moduli. The reconstruction formula involves modular inverses computed via extended GCD. Python integers handle this exactly, as do VDR integers $[n, 1, 0]$.

The decimal trap appears when CRT is used to reconstruct a rational number from its images under modular homomorphisms. Given $x \bmod p$ for several primes p , rational reconstruction recovers $x = a/b$ using extended GCD. The reconstructed fraction a/b must then be used in further computation.

10.2 The VDR Solution

The reconstructed rational a/b is represented as $[a, b, 0]$ — exact. Subsequent arithmetic on it is exact. In decimal, the reconstructed rational is truncated immediately upon entering further computation. Computer algebra systems use modular methods to compute polynomial GCDs, resultants, and determinants. The CRT reconstruction step produces exact rationals. VDR preserves them through all subsequent computation [VDR-2].

11. Partition Functions and Statistical Mechanics

11.1 The Problem

The partition function $Z = \sum \exp(-\beta E_i)$ sums Boltzmann weights over all microstates. For discrete systems, the energies E_i are often rational. The exponential is transcendental, but each Taylor partial sum is an exact rational. Free energy $F = -kT \cdot \ln(Z)$, entropy $S = -\partial F / \partial T$, specific heat $C = T \cdot \partial S / \partial T$ — all involve chains of transcendental evaluations.

The specific heat at a Schottky anomaly requires computing d^2F/dT^2 , which is a second derivative of a logarithm of a sum of exponentials — three layers of truncation in decimal. Each layer discards information that the next layer needs.

11.2 The VDR Solution

Each $\exp(-\beta E_i)$ via functional remainder Taylor series: exact rational at each depth. Sum over states: exact rational. $\ln(Z)$ via functional remainder: exact rational at each depth [VDR-1]. Discrete derivative $(F(T+h) - F(T))/h$ is exact for any rational h — the subtraction is exact (no catastrophic cancellation), and the division is exact. The Schottky specific heat peak location and height are computable to arbitrary precision by increasing Taylor depth, not by increasing decimal digits and hoping the truncation did not corrupt the answer.

11.3 Where It Matters

Phase transition analysis (specific heat divergence at critical temperature), quantum statistical mechanics (Bose-Einstein and Fermi-Dirac distributions involve $\exp(\beta E) \pm 1$ in the denominator — the ± 1 is a subtraction of a near-unit value, catastrophic in truncated decimal), astrophysics (equation of state for stellar interiors), and chemical equilibrium (equilibrium constants as ratios of partition functions).

12. Runge-Kutta with Exact Butcher Tableaux

12.1 The Problem

Runge-Kutta methods have coefficients specified as exact rationals in the Butcher tableau. Classic RK4: coefficients $1/2, 1/6, 1/3$. Higher-order methods (Dormand-Prince, Fehlberg) have coefficients like $3/40, 9/40, 12/13, 4243/5348$ — denominators with factors of 3, 7, 13, 17, 1337.

The method's order is guaranteed by algebraic identities among the coefficients: $\sum b_i = 1$, $\sum b_i c_i = 1/2$, $\sum b_i a_{ij} c_j = 1/6$. In decimal, these identities hold only approximately. If they do not hold exactly, the method is not exactly the claimed order.

12.2 The VDR Solution

Every Butcher coefficient is $[p, q, 0]$. The order conditions are verified as exact equalities: $\sum b_i = 1$ exactly, $\sum b_i c_i = 1/2$ exactly. The method is provably order 4 (or 5, or 8) because the algebraic conditions hold exactly. Each RK stage is exact rational arithmetic on exact rational data. VDR-2 gym domain 9 demonstrated: Euler method $y(1) = (11/10)^{10}$ exact, RK4 approximately $140 \times$ better accuracy than Euler — and both results are exact rationals, so the accuracy comparison is itself exact [VDR-2].

12.3 Where It Matters

Validating new RK methods (do the order conditions actually hold?), reference solutions for ODE benchmarks, and long-time integration where method error and arithmetic error must be separated. The Butcher tableau is the method's identity — if the coefficients are truncated, the method is a different method.

13. Galois Field Arithmetic and Error Correction

13.1 The Problem

Reed-Solomon codes, BCH codes, and turbo codes operate over Galois fields $GF(p^n)$. All arithmetic is modular. The syndrome computation, error locator polynomial, and Chien search involve chains of field multiplications. Some implementations in float (educational code, MATLAB scripts) silently convert field elements to doubles and round back. This works for $GF(2^8)$ (elements 0-255 fit in float mantissa) but fails for $GF(2^{16})$ and above.

13.2 The VDR Solution

$GF(p)$ elements are $[a, 1, 0]$ with a in $\{0, \dots, p-1\}$. `gf_add`, `gf_mul`, `gf_inv`, `gf_pow` builtins [VDR-10] are exact modular operations. VDR-3 gym domain 18 verified: all 6 $GF(7)$ inverses correct, all 16 Hamming(7,4) codewords have syndrome 0, all 7 single-bit errors corrected, minimum distance = 3 by exhaustive 120-pair comparison [VDR-3]. No float anywhere. The error correction is either correct or it is not — exact arithmetic means the code either corrects the error or provably does not.

13.3 Where It Matters

Satellite communication (deep space links use Reed-Solomon plus convolutional codes — error in the decoder means lost data), QR codes, data storage (RAID parity), 5G NR polar codes, and DNA storage (emerging field using Reed-Solomon over GF(4) for the ACGT alphabet).

14. Characteristic Polynomial and Cayley-Hamilton Verification

14.1 The Problem

The Cayley-Hamilton theorem states every matrix satisfies its own characteristic polynomial: $p(A) = 0$ where $p(\lambda) = \det(A - \lambda I)$. The zero is the zero matrix — every entry must be exactly zero. In decimal, $p(A)$ is a matrix with entries of order 10^{-N} where N is the working precision. The theorem cannot be verified — only approximately confirmed.

14.2 The VDR Solution

Characteristic polynomial coefficients are exact rationals computed from the exact determinant. $p(A)$ = exact zero matrix. Every entry is $[0, 1, 0]$. Not approximately zero. Zero. VDR-2 gym domain 2 and VDR-3 gym domain 21 verified: $M^2 - 5M + 5I = 0$ exactly, $A^2 + 3A + 2I = 0$ exactly [VDR-2, VDR-3]. Cayley-Hamilton is a structural identity verified by structural equality, not a numerical residual compared to a tolerance.

14.3 Where It Matters

Control theory (Cayley-Hamilton computes matrix exponentials for state-space systems), minimal polynomial computation (requires exact zero-testing to determine which powers of A are independent), and matrix function evaluation. The decimal trap makes zero-testing fundamentally unreliable: is 10^{-15} zero or nonzero? VDR answers this question exactly.

15. Farey Sequences and Stern-Brocot Trees

15.1 The Problem

The Farey sequence F_n contains all reduced fractions with denominator $\leq n$ in $[0,1]$, ordered by magnitude. The mediant property: for consecutive Farey neighbors a/b and c/d , $lad - bcl = 1$. This is an exact integer identity. Representing Farey fractions as decimals destroys the structure — you cannot recover the integer numerator and denominator from a truncated decimal without additional information.

15.2 The VDR Solution

$1/3 = [1, 3, 0]$. $1/4 = [1, 4, 0]$. Mediant $= [2, 7, 0]$. $|ad - bc| = |1 \cdot 4 - 3 \cdot 1| = 1$. Exact. VDR-2 gym domain 1 verified the mediant property for all consecutive pairs in Farey sequences [VDR-2]. The Stern-Brocot path to any rational is computable by exact comparison: VDR comparison is cross-multiplication of integers, exact and deterministic.

15.3 Where It Matters

Number theory (distribution of rationals, Ford circles, hyperbolic geometry), clock synchronization (best rational approximation to frequency ratios), and music theory (just intonation intervals as exact rationals: $3/2$ is a perfect fifth, $5/4$ is a major third, $81/80$ is the syntonic comma). Decimal representation makes the $81/80$ comma invisible — it is 1.0125 in decimal, a terminating decimal that looks unremarkable. In VDR, $81/80$ is $[81, 80, 0]$, and the relationship to $3^4/2^4 \cdot 5$ is visible in the factorization.

16. Exact Polynomial GCD and Resultants

16.1 The Problem

The GCD of two polynomials with rational coefficients is computed by the Euclidean algorithm, which involves polynomial long division producing quotients and remainders with growing denominators. The critical decision at each step: is the remainder zero? If yes, the previous divisor is the GCD. If no, the algorithm continues.

In decimal, a remainder coefficient might be 3.7×10^{-15} . Is this zero (GCD found) or nonzero (continue dividing)? Increasing precision to 50 digits might give 3.7×10^{-45} . Still ambiguous. The zero-testing problem is unsolvable in truncated decimal arithmetic — no amount of precision resolves the ambiguity without external information.

16.2 The VDR Solution

Remainder coefficient is either $[0, 1, 0]$ or it is not. Zero-testing is exact structural comparison, requiring zero digits of precision analysis. VDR-2 gym domain 2 verified: polynomial $\text{GCD}(x^2-1, x^2+2x+1) = (x+1)$ exactly [VDR-2]. The definite integral $\int_0^1 x^2 dx = 1/3$ exact via exact antiderivative evaluation (VDR-2 gym domain 13).

16.3 Where It Matters

Control theory (coprimeness of numerator and denominator polynomials determines controllability), algebraic geometry (intersection multiplicity via resultants), computer algebra (simplification of rational functions), and coding theory (polynomial GCD over finite fields for BCH decoding). In each case, the correctness of the algorithm depends on exact zero-testing that decimal arithmetic cannot provide.

17. Padé Approximants

17.1 The Problem

Computing Padé approximant coefficients requires solving a linear system where the matrix entries are Taylor coefficients — rational numbers for functions like exp, sin, cos, ln at rational arguments. Taylor coefficients involve factorials: $1/n!$ has denominators with all primes up to n . For high-order approximants ($L+M > 20$), the coefficient matrix becomes ill-conditioned and mpmath needs hundreds of digits.

$1/7! = 1/5040$. In decimal: 0.000198412698... repeating. The $[4/4]$ Padé approximant for exp(x) has coefficients with denominators up to $1680 = 2^4 \times 3 \times 5 \times 7$. In decimal, $1/1680 = 0.000595238095238...$ repeating.

17.2 The VDR Solution

$1/7! = [1, 5040, 0]$. Exact. The linear system is solved by exact Gaussian elimination or Cramer's rule. The Padé coefficients are exact rationals. The approximant's error at any rational point is computable as an exact rational — evaluate the approximant and subtract from the Taylor partial sum, both exact.

17.3 Where It Matters

Matrix exponential algorithms (Padé approximants are the standard method), signal processing (model reduction), circuit simulation (SPICE uses Padé for delay elements), and quantum field theory (Padé-Borel summation of divergent perturbation series). In each domain, the Padé coefficients involve factorials whose denominators contain odd primes, making decimal truncation unavoidable.

18. Lattice Basis Reduction

18.1 The Problem

The LLL algorithm reduces a lattice basis by iteratively performing Gram-Schmidt orthogonalization and size reduction. The Gram-Schmidt coefficients $\mu_{ij} = \langle \mathbf{b}_i, \mathbf{b}_j^* \rangle / \langle \mathbf{b}_j, \mathbf{b}_j^* \rangle$ are exact rationals when the input vectors have rational coordinates. The Lovász condition involves comparing μ^2 to the threshold $3/4$ — a comparison of exact rationals.

The critical failure mode: $\mu = 0.5000000000000001$ (above $1/2$, needs size reduction) versus $\mu = 0.4999999999999999$ (below $1/2$, does not). mpmath cannot distinguish these if the true value is within truncation range of $1/2$. Wrong decision means wrong basis, wrong lattice reduction, wrong cryptanalysis result.

18.2 The VDR Solution

$\mu = [p, q, 0]$ exact rational. Comparison to $1/2$: cross-multiply $p \cdot 2$ versus $q \cdot 1$, compare exact integers. No borderline ambiguity. The Lovász condition is an exact rational comparison. VDR-3 gym domain 20 verified: lattice $\mu_{21} = 1/2$ exact, $v_1 \cdot v_2^* = 0$ exactly, Lovász condition checked with exact rational comparison [VDR-3].

18.3 Where It Matters

Cryptanalysis (breaking lattice-based cryptosystems requires correct basis reduction), integer programming (finding short vectors), number theory (finding algebraic relations between constants), and GPS integer ambiguity resolution. In each case, the algorithm's correctness depends on exact threshold comparisons that decimal truncation makes unreliable.

19. Groebner Bases

19.1 The Problem

Groebner basis computation (Buchberger's algorithm) involves polynomial division with remainder over multivariate polynomials. Each S-polynomial computation and reduction step produces polynomials with growing denominators. The critical decision at each step: is the remainder zero?

This is the same zero-testing problem as polynomial GCD (Section 16), but amplified. Groebner basis computation involves many more reduction steps, each producing larger denominators. A remainder coefficient of 10^{-15} could be zero (reduction succeeds, S-polynomial processed) or nonzero (new basis element needed). Wrong answer means wrong Groebner basis — extra elements, missing elements, or both.

19.2 The VDR Solution

Remainder coefficient is either $[0, 1, 0]$ or it is not. Zero-testing is exact. The Groebner basis computation terminates with the correct basis — not an approximately correct basis that might have extra or missing elements.

19.3 Where It Matters

Algebraic geometry (ideal membership testing, variety computation), robotics (solving polynomial systems for inverse kinematics), coding theory (decoding algebraic codes), and algebraic statistics (design of experiments). Each application depends on the Groebner basis being correct, not approximately correct.

20. Quantum Error Correction Stabilizer Codes

20.1 The Problem

Stabilizer codes are defined by Pauli group elements — tensor products of I, X, Y, Z matrices. The entries are from $\{0, \pm 1, \pm i\}$. Checking code properties requires multiplying Pauli matrices and determining whether results are $\pm I$ (real) or $\pm iI$ (imaginary).

In float, i is represented as $0 + 1.0j$. Multiplying: $(0+1j) \cdot (0+1j) = -1+0j$ in exact arithmetic, but float produces $-1+1.2 \times 10^{-16}j$. After 20 matrix multiplications, the imaginary part of what should be a real number is approximately 10^{-15} . Is the result $\pm I$ (real) or $\pm iI$ (imaginary)? Float cannot determine this for borderline cases.

20.2 The VDR Solution

i is represented as a pair of VDR values $(0, 1)$. Multiplication: $(0+1i) \cdot (0+1i) = (-1, 0)$. Exact. After 20 multiplications: exact. The result is either $(\pm 1, 0)$ — real, meaning $\pm I$ — or $(0, \pm 1)$ — imaginary, meaning $\pm iI$. Exact classification. VDR-13 verified Pauli algebra: $\sigma_x^2 = I$, $\sigma_x \cdot \sigma_y = i\sigma_z$ as exact structural equalities [VDR-13].

20.3 Where It Matters

Designing quantum error correcting codes, verifying fault-tolerance thresholds, topological code analysis (surface codes, color codes), and magic state distillation protocols. Code distance computation requires checking if a Pauli operator commutes with all stabilizers — commutation is exact comparison of matrix products. Decimal truncation makes this comparison unreliable for products of many Pauli matrices.

21. The Unifying Principle

Every domain in this paper contains the same structural vulnerability: a computation chain where intermediate values have denominators with prime factors other than 2 and 5, forcing decimal or mpmath truncation at every step.

$1/3$ is infinite in decimal. $1/7$ is infinite. $1/11$ is infinite. $1/5040$ is infinite ($5040 = 2^4 \times 3^2 \times 5 \times 7$, the factors of 3 and 7 make it repeat). Every factorial, every prime denominator, every product of odd primes produces an infinite repeating decimal.

VDR's representation $[V, D, 0]$ carries the exact denominator as an integer — whether it is 3, 7, 13, 5040, 117117, or a 200-digit product of primes. No truncation. No repeating digits. No lost information. The denominator grows but nothing is discarded.

This is not “more precision.” It is a categorically different relationship with the numbers. mpmath says “I have 1000 digits, which is probably enough.” VDR says “I have the exact answer.”

22. Boundaries

22.1 Denominator Growth

Exact rational arithmetic preserves every prime factor in every denominator. After N multiplications, the denominator can have $O(N)$ distinct prime factors and magnitude growing multiplicatively. For long chains, the integers become very large. Q335 fixed-frame arithmetic [VDR-14, VDR-18] bounds the denominator at 2^{335} and carries overflow in the Remainder slot.

22.2 Transcendental Functions

Functions like $\exp$, $\sin$, $\cos$, $\log$ at irrational arguments produce irrational results. VDR handles these through functional remainders — Newton iteration or Taylor series producing exact rational approximations at each depth [VDR-1]. The approximation is configurable, inspectable, and does not truncate — the Remainder carries exactly what the current depth did not absorb.

22.3 Matrix Size

Exact matrix operations via cofactor expansion are $O(n!)$. Gaussian elimination is $O(n^3)$ but requires pivot selection. Practical exact matrix operations are currently limited to approximately 50×50 . For larger systems, VDR serves as ground truth for validating float implementations.

22.4 Not All Domains Benefit Equally

Domains where all intermediate values have denominators of the form $2^m \times 5^n$ — pure decimal-compatible arithmetic — do not benefit from VDR's rational representation. Binary arithmetic (denominators that are powers of 2) is already exact in float. VDR's value is proportional to the density of odd prime factors in the computation's intermediate denominators.

Appendices

Appendix A — Denominator Prime Factor Analysis by Domain

Domain	Typical denominator prime factors	Decimal period	mpmath digits needed for N -step chain	VDR digits needed
Continued fractions	All primes up to convergent index	Variable, often large	$O(N \times \text{max period})$	0 (exact)
Bernoulli numbers	All primes p where $(p-1)$ divides $2n$	Up to thousands (large primes)	$O(\text{period} \times \text{chain length})$	0 (exact)

Domain	Typical denominator prime factors	Decimal period	mpmath digits needed for N-step chain	VDR digits needed
Hilbert matrix	All primes up to $2n-1$	LCM of all primes	Grows with condition number $\times n$	0 (exact)
Eigenvalue (2×2 rational)	Factors of trace and determinant	Product of all factors	Depends on discriminant	0 (exact)
Gaussian elimination pivots	$O(n)$ distinct primes	Product of all pivot factor periods	$O(n^2 \times \text{max period})$	0 (exact)
Bayesian updates	Factors of all priors and likelihoods	Cumulative product of periods	$O(N \times \text{cumulative period})$	0 (exact)
Markov steady state	All factors from transition matrix	Product of all periods	$O(k^2 \times \text{max period})$	0 (exact)
Elliptic curve points	Height growth: $O(4^n)$ digits	Enormous	$O(4^n)$	0 (exact, but integers are huge)
RK Butcher tableau	All primes in coefficient denominators	Product of periods	$O(\text{stages} \times \text{max period})$	0 (exact)
Galois field	p (field characteristic)	N/A (modular)	N/A	0 (exact modular)
Polynomial GCD	All primes from coefficient denominators	Product of periods	Unknown (zero-testing problem)	0 (exact zero-testing)
Padé coefficients	Primes up to $(L+M)!$	Product of periods	$O(\text{condition number} \times \text{matrix size})$	0 (exact)
Lattice LLL	Factors from Gram-Schmidt coefficients	Product of periods	$O(n^2 \times \text{iterations} \times \text{max period})$	0 (exact)
Groebner basis	All primes from polynomial coefficients	Product of periods	Unknown (zero-testing problem)	0 (exact zero-testing)
Partition function	Primes from energy denominators	Product of periods	$O(\text{states} \times \text{Taylor depth} \times \text{max period})$	0 (exact per Taylor depth)

Domain	Typical denominator prime factors	Decimal period	mpmath digits needed for N-step chain	VDR digits needed
Stern-Brocot	All primes (every rational appears)	Variable	N/A (structure destroyed by decimal)	0 (exact; structure preserved)
Quantum stabilizer	Only 1 (entries are $\pm 1, \pm i$)	N/A	N/A (but complex arithmetic drifts)	0 (exact complex pairs)
Cayley-Hamilton	All primes from matrix entries	Product of periods	$O(n \times \text{condition number} \times \text{max period})$	0 (exact zero-testing)

Appendix B — Zero-Testing Failure Modes

Domain	Zero-testing question	Decimal behavior	VDR behavior	Consequence of wrong answer
Polynomial GCD	Is remainder coefficient zero?	Ambiguous: could be truncation artifact	Exact: $[0,1,0]$ or not	Wrong GCD: extra factors or missing factors
Groebner basis	Is S-polynomial remainder zero?	Ambiguous at any precision	Exact	Wrong basis: extra or missing generators
Cayley-Hamilton	Is $p(A)$ the zero matrix?	Every entry $\approx 10^{-N}$, never exactly 0	Every entry exactly $[0,1,0]$	Cannot verify fundamental theorem
Matrix rank	Is singular value zero or small?	Indistinguishable below $\sim 10^{-15}$	Exact zero or exact nonzero	Wrong rank: wrong nullspace dimension
Linear independence	Is determinant zero?	Ambiguous near zero	Exact	Wrong independence conclusion
Eigenvector orthogonality	Is dot product zero?	$\approx 10^{-15}$, never exactly 0	Exactly 0 (for rational eigenvectors)	Cannot verify orthogonality
Polynomial coprimeness	Is $\text{GCD} = 1$?	Requires zero-testing of remainders	Exact	Wrong controllability assessment

Domain	Zero-testing question	Decimal behavior	VDR behavior	Consequence of wrong answer
Lattice LLL threshold	Is $\mu > 1/2$?	Ambiguous near $1/2$	Exact comparison	Wrong basis reduction
Farey mediant property	Is	ad-bc	$= 1$?	Requires exact integer recovery from decimal
Conservation law	Is energy change zero?	$\approx 10^{-15}$, declared “conserved to tolerance”	Exactly 0 or exactly nonzero	False conservation or missed violation

Appendix C — Repeating Decimal Periods for Common Denominators

Denominator	Prime factorization	Decimal period	Representation in decimal	Representation in VDR
2	2	0 (terminates)	0.5	[1, 2, 0]
3	3	1	0.333...	[1, 3, 0]
4	2^2	0	0.25	[1, 4, 0]
5	5	0	0.2	[1, 5, 0]
6	$2 \cdot 3$	1	0.1666...	[1, 6, 0]
7	7	6	0.142857142857...	[1, 7, 0]
8	2^3	0	0.125	[1, 8, 0]
9	3^2	1	0.111...	[1, 9, 0]
10	$2 \cdot 5$	0	0.1	[1, 10, 0]
11	11	2	0.090909...	[1, 11, 0]
12	$2^2 \cdot 3$	1	0.08333...	[1, 12, 0]
13	13	6	0.076923076923...	[1, 13, 0]
14	$2 \cdot 7$	6	0.0714285714285...	[1, 14, 0]
15	$3 \cdot 5$	1	0.0666...	[1, 15, 0]
17	17	16	0.05882352941176...	[1, 17, 0]
19	19	18	0.052631578947368421...	[1, 19, 0]
21	$3 \cdot 7$	6	0.047619047619...	[1, 21, 0]
100	$2^2 \cdot 5^2$	0	0.01	[1, 100, 0]
360	$2^3 \cdot 3^2 \cdot 5$	1	0.00277...	[1, 360, 0]
2730	$2 \cdot 3 \cdot 5 \cdot 7 \cdot 13$	12 (lcm of 1,6,6 for 3,7,13)	0.000366300... repeating	[1, 2730, 0]
5040	$2^4 \cdot 3^2 \cdot 5 \cdot 7$	6	0.000198412698... repeating	[1, 5040, 0]
117117	$3 \cdot 7 \cdot 11 \cdot 13 \cdot 37$ (approx)	$\text{lcm}(1,6,2,6,\dots)$	Long repeating block	[1, 117117, 0]

Of the first 20 denominators, 8 terminate (40%) and 12 repeat (60%). Among denominators from 1 to 100, approximately 12 terminate and 88 repeat. The decimal trap affects the vast majority of rational numbers.

Appendix D — Prior VDR Validation by Domain

Section	Domain	VDR paper	Gym domain	Tests	Key result
2	Continued fractions	VDR-2	3	19/19	Gauss-Kuzmin statistics exact
3	Bernoulli/zeta	VDR-3	Q335	157 (152 pass)	22 constants at 100 digits
4	Hilbert matrix	VDR-1	Core	68/68	H^{-1} residual = 0; $\det(H_4)$ = 1/6048000 exact
5	Eigenvalues	VDR-2	4, 11	38/37 (1 test error)	Exact steady state [2/3, 1/3]
6	Gaussian elimination	VDR-13	Physical	—	H_{30} pivots: 185-digit denominators, zero error
7	Bayesian updates	VDR-2	11	19/19	Posterior 6/7 exact; binomial PMF sums to 1
8	Markov chains	VDR-2, VDR-3	11, 16	38/38	PageRank sums to exactly 1
9	Elliptic curves	—	—	—	Structural analysis; not yet in gym
10	CRT/modular	VDR-2	12	19/18 (1 test error)	RSA roundtrip, CRT exact

Section	Domain	VDR paper	Gym domain	Tests	Key result
11	Partition functions	—	—	—	Taylor via functional remainder validated
12	Runge-Kutta	VDR-2	9	19/18 (1 test error)	RK4 140 $\times$ better than Euler; both exact
13	Galois fields	VDR-3	18	—	Hamming(7,4): all errors corrected
14	Cayley-Hamilton	VDR-2, VDR-3	2, 21	—	$M^2 - 5M + 5I = 0$ exactly
15	Farey/Stern-Brocot	VDR-2	1	19/19	Mediant property verified
16	Polynomial GCD	VDR-2	2	18/18	$\text{GCD}(x^2-1, x^2+2x+1) = (x+1)$ exact
17	Padé approximants	—	—	—	Linear system solve via exact elimination
18	Lattice LLL	VDR-3	20	—	$\mu_{21} = 1/2$ exact; Lovász exact
19	Groebner bases	—	—	—	Structural analysis; zero-testing exact
20	Quantum stabilizer	VDR-13	Physical	—	Pauli algebra: $\sigma^2 = I$ exact
Total	20 domains	6 papers	14 gym domains	>400 tests	Zero VDR computation errors

Appendix E — Catastrophic Cancellation Vulnerability by Domain

Domain	Cancellation operation	Operand magnitudes	Decimal digits lost	VDR digits lost
Continued fraction quality	$\alpha - p/q$	Both $\approx$ irrational value	Most significant digits cancel	0
Eigenvalue near zero	$(a+d)/2 - \sqrt{\text{disc}}$	$\text{Trace}/2 \approx \sqrt{\text{discriminant}}$	Up to all digits	0
Specific heat (Schottky)	$F(T+h) - F(T)$	Both $\approx F(T)$	Proportional to $-\log_{10}(h)$	0
Options Greeks	$f(S+h) - f(S-h)$	Both $\approx f(S)$	Proportional to $-\log_{10}(h)$	0
Bose-Einstein denominator	$\exp(\beta E) - 1$	$\exp(\beta E) \approx 1$ for small βE	Most digits	0
Stability eigenvalue	Real part of complex eigenvalue	Real ≈ 0	Most digits	0 (to Newton depth)
Cayley-Hamilton verification	$A^n - c_1 A^{n-1} - \dots$	Terms of similar magnitude	Many digits per entry	0
Gaussian elim small pivot	$a - (b \cdot c/d)$	$a \approx b \cdot c/d$	Proportional to condition number	0
Kalman P update	$(I-KH)P$ entries	$(I-KH) \approx I$ for good observations	Up to several digits	0
Polynomial GCD remainder	$p(x) - q(x) \cdot \text{quotient}(x)$	Nearly equal polynomials	May lose all significant digits	0
LLL μ near 1/2	$\mu - 1/2$	$\mu \approx 1/2$	May lose all significant digits	0

Appendix F — Cost of Exactness by Domain

Domain	Typical problem size	Float64 operations	VDR operations (Python)	VDR/Float ratio	Justified by
Continued fraction (200 convergents)	200 iterations	~ 600	$\sim 30,000$	$50 \times$	Exact quality measures
Hilbert inverse (10×10)	10^3 multiplications	$\sim 1,000$	$\sim 50,000$	$50 \times$	Zero vs 10^{-20} residual

Domain	Typical problem size	Float64 operations	VDR operations (Python)	VDR/Float ratio	Justified by
Bayesian (100 updates)	$100 \times \sim 10$ operations	$\sim 1,000$	$\sim 50,000$	$50 \times$	Exact posterior for diagnostics
Markov steady (5 states, power method)	$5^2 \times 100$ iterations	$\sim 2,500$	$\sim 125,000$	$50 \times$	Exact PageRank
RK4 (100 steps, 1D)	$100 \times \sim 10$ operations	$\sim 1,000$	$\sim 50,000$	$50 \times$	Provable method order
Polynomial GCD (degree 10)	~ 100 operations	~ 100	$\sim 5,000$	$50 \times$	Exact zero-testing
LLL (dimension 5)	~ 100 iterations $\times$ ~ 25 ops	$\sim 2,500$	$\sim 125,000$	$50 \times$	Exact threshold comparison
Eigenvalue (2×2)	~ 10 operations	~ 10	~ 500	$50 \times$	Exact stability determination
Elliptic curve (20 doublings)	~ 200 operations	~ 200	$\sim 10,000$	$50 \times$	Exact curve membership
Hamming(7,4) decode	~ 50 operations	~ 50	$\sim 2,500$	$50 \times$	Exact error correction
Cayley-Hamilton (5×5)	$5^5 + 5^4 + \dots$ operations	$\sim 4,000$	$\sim 200,000$	$50 \times$	Exact zero verification
Groebner (3 variables, degree 3)	~ 500 operations	~ 500	$\sim 25,000$	$50 \times$	Exact zero-testing
Padé [4/4]	~ 100 operations (linear solve)	~ 100	$\sim 5,000$	$50 \times$	Exact coefficients
Partition function (10 states)	$10 \times \sim 30$ Taylor terms	~ 300	$\sim 15,000$	$50 \times$	Exact thermodynamics
Quantum stabilizer (5 qubits)	~ 100 Pauli products	~ 100	$\sim 5,000$	$50 \times$	Exact commutation check

The ratio is approximately $50 \times$ in Python across all domains. On GPU with Q335, the ratio is approximately $150 \times$ per operation but offset by the elimination of multiple-precision overhead and the ability to parallelize fixed-width operations [VDR-18].

Appendix G — The Decimal Compatibility Test

A computation is decimal-compatible if every intermediate value has a denominator whose only prime factors are 2 and 5. For any computation, the following test determines whether the decimal trap applies:

Input property	Decimal-compatible?	VDR advantage
All inputs are integers	Yes (denominator = 1)	None for addition/subtraction; multiplication stays integer
All inputs are powers of 2 in denominator	Yes	None
All inputs are powers of 10 in denominator	Yes	None
Any input has factor of 3 in denominator	No	Exact vs truncated
Any input has factor of 7 in denominator	No	Exact vs truncated
Any input has any odd prime > 5 in denominator	No	Exact vs truncated
Division by 3 anywhere in chain	No (introduces factor of 3)	Exact vs truncated
Division by any odd number > 5	No	Exact vs truncated
Mean of N values where N has odd factor	No (introduces $1/N$)	Exact vs truncated
Factorial in denominator ($n!$ for $n \geq 3$)	No (3 divides $3!$)	Exact vs truncated
Any prime modulus (Galois field)	No (p in denominator of inverses)	Exact modular vs float approximation
Bayesian normalization	Almost always no	Exact vs truncated
Matrix entries with odd denominators	No	Exact linear algebra

The test is simple: if division by anything other than a power of 2 or 5 occurs anywhere in the computation — including implicitly through averaging, normalization, matrix inversion, or polynomial evaluation — the decimal trap applies and VDR provides exact results where decimal provides truncated approximations.

Appendix H — Domain Interconnection Map

Domain A	Domain B	Shared vulnerability	Example
Bernoulli numbers	Padé approximants	Factorial denominators with odd primes	exp Padé coefficients involve $1/n!$
Hilbert matrix	Gaussian elimination	Growing pivot denominators	H inverse requires exact pivots
Eigenvalues	Stability analysis	Discriminant with odd factors	Eigenvalue sign from exact discriminant
Eigenvalues	Markov chains	Steady state is eigenvector for $\lambda=1$	Steady state denominators from transition matrix
Bayesian updates	Medical diagnostics	Sequential rational multiplication	Posterior denominators grow with evidence
Polynomial GCD	Control theory	Coprime-ness requires zero-testing	Controllability from exact polynomial GCD
Polynomial GCD	Groebner bases	Both depend on exact zero-testing	Remainder zero $\rightarrow$ different algorithm path
Galois fields	Error correction	Exact modular chains	Syndrome computation, error locator
Galois fields	Cryptography	Exact field arithmetic	RSA, ECC, post-quantum lattice
Lattice LLL	Cryptanalysis	Exact threshold comparison	Breaking lattice-based encryption
Lattice LLL	GPS	Integer ambiguity resolution	Carrier phase positioning
Elliptic curves	Cryptography	Exact point arithmetic	ECC key generation, ECDSA
Runge-Kutta	Physics simulation	Exact Butcher coefficients	ODE integration with proven order
Cayley-Hamilton	Control theory	Exact matrix polynomial evaluation	Matrix exponential via CH
Partition function	Quantum mechanics	Exact Boltzmann weights	Energy level statistics
Continued fractions	Number theory	Exact convergent arithmetic	Irrationality measures

Domain A	Domain B	Shared vulnerability	Example
Farey sequences	Number theory	Exact mediant property	Rational approximation theory
Quantum stabilizer	Quantum computing	Exact Pauli algebra	Error correction code design
Markov chains	PageRank	Exact steady state	Web ranking
Gaussian elimination	All linear algebra	Exact pivots	Foundation for inverse, rank, determinant

Appendix I — Additional Domains: Algebraic Number Theory

Problem	Computation	Decimal failure	VDR solution	Verified
Minimal polynomial of $\sqrt{2}+\sqrt{3}$	x^4-10x^2+1 ; verify $(\sqrt{2}+\sqrt{3})$ is root	Substituting truncated $\sqrt{2}+\sqrt{3}$ gives residual $\sim 10^{-15}$	Newton $\sqrt{2} + \sqrt{3}$ substituted; residual exact at Newton depth	VDR-1 functional remainder
Norm of algebraic integer	$N(a+b\sqrt{d}) = a^2-db^2$	$a=3/7, b=5/11, d=2$: exact rational	$[9/49 - 50/121, 1, 0] = [9 \cdot 121 - 50 \cdot 49 / 49 \cdot 121, 1, 0] = [-1391/5929, 1, 0]$ exact	VDR-1 closed arithmetic
Discriminant of number field	$\Delta(Q(\sqrt{d})) = 4d$ or d depending on $d \bmod 4$	$d=13$: $\Delta=13$ exact integer	$[13, 1, 0]$	Trivial
Class number computation	Count ideal classes via Minkowski bound	Bound involves $\sqrt{\Delta}$; class group enumeration involves exact norm comparisons	$\sqrt{13}$ via Newton; norms as exact rationals; comparisons exact	VDR-2 gym 1

Problem	Computation	Decimal failure	VDR solution	Verified
Dedekind zeta residue	Residue involves class number $\times$ regulator $\times 2^{r_1} \times (2\pi)^{r_2} / (w \cdot \sqrt{D})$	Δ	)	Product of transcendentals and exact rationals; truncation at every factor
Pell equation	$x^2 - Dy^2 = 1$; fundamental solution from continued fraction of $\sqrt{D}$	CF convergents exact; checking $x^2 - Dy^2 = 1$ requires exact large integer arithmetic	Convergents $[p_n, q_n, 0]$; verification $p_n^2 - D \cdot q_n^2 = 1$ exact	VDR-2 gym 3
Unit group of $\mathbb{Z}[\sqrt{d}]$	Fundamental unit $\varepsilon = (a + b\sqrt{d})/c$	Exact rational coefficients with Newton $\sqrt{d}$; unit equation $N(\varepsilon) = \pm 1$ verified exactly	VDR-1 closed + functional	
Resultant of cyclotomic polynomials	$\text{Res}(\Phi_m, \Phi_n) = p^{\varphi(m)}$ if $n/m = p^k$, else 1	Coefficients are ± 1 ; resultant is Sylvester determinant of exact integers	$[p^{\varphi(m)}, 1, 0]$ exact integer	VDR-2 gym 2

Appendix J — Additional Domains: Combinatorial Optimization

Problem	Computation	Decimal failure	VDR solution	Key operation
Linear programming simplex	Pivot ratios $\min(b_i / a_{ij})$ for rational data	Pivots truncated; wrong pivot selection possible near ties	Exact rational pivot ratios; exact min comparison	Cross-multiply comparison
Integer programming relaxation bound	LP relaxation value as exact rational	Truncated bound may be above/below true bound	Exact rational bound; integrality gap exact	Exact simplex

Problem	Computation	Decimal failure	VDR solution	Key operation
Assignment problem (Hungarian)	Subtract row/column minima; find augmenting paths	Rational costs truncated; path selection affected near ties	Exact costs; exact path selection	Exact subtraction and comparison
Network flow max-flow	Augmenting path capacities; residual graph	Rational capacities truncated; residual near zero ambiguous	Exact capacity subtraction; zero-flow detection exact	Exact zero-testing
Knapsack fractional	Greedy by value/weight ratio	Ratios like 7/13 truncated; wrong ordering possible near ties	[7, 13, 0] vs [5, 9, 0]; exact cross-multiply comparison	Exact comparison
Traveling salesman (branch and bound)	Lower bound from LP relaxation	Truncated bounds cause wrong pruning	Exact bound; pruning decisions exact	Exact comparison
Minimum spanning tree (Kruskal)	Sort edges by weight; union-find	Rational weights truncated; wrong sort order near ties	Exact sort by cross-multiplication	Exact comparison
Matroid intersection	Augmenting path in exchange graph	Weight comparisons on rational edge values	Exact rational comparison	Cross-multiply

Appendix K — Additional Domains: Algebraic Geometry

Problem	Computation	Decimal failure	VDR solution	Key property
Bézout's theorem verification	Intersection multiplicity = product of degrees	Compute resultant of two curves; check degree of resultant	Exact resultant via exact Sylvester determinant	Exact integer determinant

Problem	Computation	Decimal failure	VDR solution	Key property
Genus of algebraic curve	$g = (d-1)(d-2)/2$ for smooth degree- d curve	Integer formula; but verifying smoothness requires exact discriminant	Discriminant as exact rational; zero-testing exact	Exact zero-testing
Rational points on curves	Find $(x,y) \in \mathbb{Q}^2$ satisfying $f(x,y)=0$	Evaluate f at candidate rationals; truncation makes zero-testing impossible	$f(p/q, r/s)$ evaluated as exact rational; is it $[0,1,0]$?	Exact evaluation
Divisor class group	Compute divisors of rational functions; check linear equivalence	Function evaluation at rational points truncated	Exact evaluation; exact comparison of divisors	Exact arithmetic chain
Intersection form	Bilinear form on homology; matrix of intersection numbers	Integer matrix; but computation involves exact topology $\rightarrow$ algebra chain	Exact integer matrix operations	Exact linear algebra
Hilbert function computation	$\dim(R/I)$ for graded ring R and ideal I	Requires exact Groebner basis (Section 19 of paper)	Exact Groebner $\rightarrow$ exact dimension count	Exact zero-testing
Blow-up computation	Local coordinates with rational transformation matrices	Transformation matrices have rational entries; chains of transforms	Exact matrix multiplication chain	Exact rational chain

Appendix L — Additional Domains: Representation Theory

Problem	Computation	Decimal failure	VDR solution	Verified
Character table of finite group	Characters are algebraic integers; for abelian groups, roots of unity	Character values involve 1/	G	normalization;
Schur orthogonality	$\sum \chi_i(g) \cdot \chi_j(g)^*$ =	G	$\cdot \delta_{i,j}$	Sum of products of ratio-nals/algebraic integers; truncation breaks orthogonality
Decomposition into irreducibles	Multiplicity = (1/	G	$) \sum \chi(g) \cdot \chi_i(g)^*$	Division by
Young tableaux hook lengths	Frame-Robinson-Thrall: $\dim = n! / \prod \text{hook lengths}$	Hook lengths are small integers; product has many odd primes	Exact integer product; $\dim$ is exact integer	VDR-2 gym 5
Tensor product decomposition	Clebsch-Gordan coefficients as exact rationals	Coefficients involve square roots of factorials divided by products	Newton sqrt of exact rationals; exact chains	VDR-1 functional remainder
Burnside's lemma		X/G	= (1/	G
Molien series	1/	G	$\sum 1/\det(I-gT)$	Each determinant is polynomial in T with rational coefficients; sum involves 1/

Appendix M — Additional Domains: Topology and Homology

Problem	Computation	Decimal failure	VDR solution	Key property
Smith normal form	Diagonalize integer matrix over $\mathbb{Z}$	Integer row/column operations; intermediate entries grow	Exact integer arithmetic throughout; SNF entries are exact divisors	Exact integer chain
Homology group computation	Kernel/image of boundary matrices	Rank computation requires exact zero-testing	Exact rank via exact elimination	Exact zero-testing
Betti numbers	$\beta_k = \text{rank}(H_k) = \text{rank}(\ker \partial_k) - \text{rank}(\text{im } \partial_{k+1})$	Exact integer rank difference	Both ranks exact; difference exact	Exact linear algebra
Euler characteristic	$\chi = \sum (-1)^k \beta_k$	Alternating sum of exact integers	Exact integer sum	Trivial once ranks exact
Intersection number	Algebraic count of intersection points	Integer by definition; but computation goes through exact matrix ops	Exact determinant/rank computation	Exact linear algebra
Fundamental group presentation	Word problem in finitely presented groups	Involves exact equality of group elements	Exact comparison of normal forms	Exact structural equality
Simplicial chain complex	Boundary operator as integer matrix	All entries ± 1 or 0; but products have growing entries	Exact integer matrix product	Exact arithmetic
Persistent homology	Track birth/death of features across filtration parameter	Filtration parameter comparisons; birth/death times as rationals	Exact comparison of rational filtration values	Exact comparison

Appendix N — Additional Domains: Category Theory and Logic

Problem	Computation	Decimal failure	VDR solution	Key property
Functor composition verification	$F \circ G \circ H$ applied to morphisms	Morphisms as rational matrices; composition is matrix multiply	Exact matrix multiply chain	Exact rational chain
Natural transformation coherence	Square commutes: $\eta \circ B \circ F(f) = G(f) \circ \eta \circ A$	Product of exact rational matrices; equality check	Both paths produce exact matrix; structural equality	Exact zero-testing
Kan extension computation	Colimit formula involving products and coproducts	Rational coefficients in colimit cone	Exact rational coefficients; universal property verified exactly	Exact comparison
Topos-theoretic truth values	Subobject classifier Ω ; characteristic morphisms	For finite categories: rational-valued truth	Exact rational truth values; conjunction/disjunction exact	Exact arithmetic
Boolean satisfiability with weighted clauses	Max-SAT with rational clause weights	Weight sums truncated; optimal may be misidentified	Exact rational weight sums; exact comparison	Exact comparison
Proof verification	Type-checking terms in dependent type theory	Definitional equality requires exact comparison	VDR values in types compared exactly	Exact structural equality

Appendix O — Additional Domains: Information Theory

Problem	Computation	Decimal failure	VDR solution	Key operation
Shannon entropy $H = -\sum p_i \log(p_i)$	Rational probabilities, transcendental log	log via functional remainder; product with exact p_i	$[p_i, 1, 0] \times \log([p_i, D, 0])$ via Taylor	VDR-1 functional remainder

Problem	Computation	Decimal failure	VDR solution	Key operation
Mutual information $I(X;Y) = H(X) + H(Y) - H(X,Y)$	Subtraction of nearly-equal entropies	Catastrophic cancellation when X,Y nearly independent	Exact subtraction of Taylor-evaluated logs	Exact subtraction
KL divergence $D \text{ KL}(P\ Q) = \sum p_i \log(p_i / q_i)$	Rational P, Q ; log of rational ratio	$\log(p_i / q_i)$ truncated; sum accumulates	$\log([p_i \cdot q_j \text{ denom}, q_i \cdot p_j \text{ denom}, 0])$ via Taylor; exact sum	Exact rational argument to Taylor
Channel capacity	$\max \{p(x)\}$ $I(X;Y)$ subject to $\sum p_i = 1$	Optimization over rational probabilities	Lagrange conditions with exact rational arithmetic	Exact optimization
Rate-distortion function	$\min \{p(\hat{x} \mid x)\}$ $I(X;\hat{X})$ subject to $E[d(X,\hat{X})] \leq D$		Blahut-Arimoto iteration with rational updates	Each iteration exact; convergence to exact fixed point
Kolmogorov complexity bounds	Counting argument over exact string lengths	Integer arithmetic	Exact integer	Trivial
Arithmetic coding interval	Subdivide $[0,1)$ by exact cumulative probabilities	Cumulative sums of $1/3, 1/7$, etc. truncated; interval boundaries wrong	Exact cumulative sums; interval boundaries exact rationals	Exact rational sum
Lempel-Ziv dictionary sizes	Integer counts and ratios	Compression ratio as exact rational count/original	Exact integer division	Exact ratio

Appendix P — Additional Domains: Differential Geometry (Discrete/Rational)

Problem	Computation	Decimal failure	VDR solution	Key operation
Discrete curvature on mesh	Angle defect = $2\pi - \sum \theta_i$ at vertex	Angles via arctan of rational edge ratios; sum truncated	arctan via Taylor functional remainder; exact sum	Exact sum of Taylor values
Christoffel symbols (rational metric)	$\Gamma^i_{j\ k} = (1/2)g^{il}(\partial g_{j\ l} / \partial x^k + \partial g_{k\ l} / \partial x^j - \partial g_{j\ k} / \partial x^l)$	Metric entries rational; discrete derivatives exact; inverse metric exact	All operations on exact rationals; Γ entries exact	Exact matrix inverse
Geodesic equation (discrete)	$\ddot{x}^i + \Gamma^i_{j\ k} \dot{x}^j \dot{x}^k = 0$; step with exact Γ	Exact Christoffel $\times$ exact velocity; Euler step exact	$[\ddot{x}^i, 1, 0]$ exact; position update exact	Exact multiplication chain
Riemann tensor components	$R^i_{j\ k\ l}$ from derivatives of Γ	Second derivatives of metric; products of Γ values	Exact discrete derivatives; exact products	Exact rational chain
Euler characteristic via Gauss-Bonnet	$\chi = (1/2\pi) \int K\ dA \rightarrow$ discrete sum	Discrete curvatures summed; π factor via Q335	Exact discrete curvatures; Q335 π ; exact sum	Exact sum $\times$ Q335
Parallel transport (discrete)	Rotate vector by connection form at each edge	Rotation by exact rational angle (discrete); product of rotation matrices	Each rotation matrix exact (rational trig approximation); product exact	Exact matrix product chain
Holonomy computation	Product of parallel transport around loop	Chain of rotation matrices; should return to identity for flat	Exact matrix product; compare to identity exactly	Exact identity check

Appendix Q — Additional Domains: Mathematical Physics

Problem	Computation	Decimal failure	VDR solution	Prior validation
Clebsch-Gordan coefficients	$C^{j_3 m_3}_{j_1 m_1, j_2 m_2}$ as exact rationals involving factorials	Factorial ratios with many odd prime factors	Exact: factorials as integers, ratios as exact rationals	VDR-2 gym 5
Wigner 3j/6j/9j symbols	Sums of products of factorial ratios	Same as Clebsch-Gordan; longer chains	Exact: every intermediate product exact	VDR-2 gym 5
Racah coefficients	Linear combinations of 3j symbols	Sum of exact rationals; no truncation	Exact sum of exact rationals	VDR-1
Spherical harmonics at rational angles	$Y_l^m(\theta, \varphi)$ involves associated Legendre polynomials (rational for rational $\cos \theta$) $\times \exp(im\varphi)$	Polynomial part truncated if $\cos \theta$ has odd denominator	Legendre polynomial evaluated exactly at rational $\cos \theta$	VDR-2 gym 2
Partition function (Ising model)	$Z = \sum \exp(-\beta J \sum s_i s_j)$ over spin configurations	Each exponential truncated; sum of truncations	Taylor exp at each configuration; exact sum	VDR-1 functional remainder
Transfer matrix eigenvalues	$2^s \times 2^s$ matrix for s-site strip; eigenvalues determine free energy	Matrix entries involve $\exp(-\beta J)$; eigenvalue computation on truncated matrix	Exact Taylor entries; exact eigenvalue (2×2) or Newton (larger)	VDR-2 gym 4
Casimir energy	$\zeta(-1) = -1/12$ (Ramanujan summation)	Exact rational	$[-1, 12, 0]$ exact	VDR-3 Q335
Fine structure constant contributions	$\alpha = e^2/(4\pi\epsilon_0\hbar c)$; QED corrections as rational combinations of π , $\zeta(3)$, $\ln(2)$	Each correction term is rational $\times$ Q335 constant	Exact rational $\times$ Q335 integer; sum in shared frame	VDR-13 QED

Problem	Computation	Decimal failure	VDR solution	Prior validation
Bethe ansatz momenta	Solve coupled rational equations for momentum parameters	Each equation involves rational functions of momenta	Exact rational equation solving	Exact polynomial arithmetic

Appendix R — Additional Domains: Numerical Methods Theory

Problem	Computation	Decimal failure	VDR solution	Key insight
Convergence order verification	Ratio $e_{n+1}/e_n \rightarrow C$; determine p	Errors e_n computed as $f(x_n) - f(x^*)$; subtraction of nearly-equal values	Exact error: $[f(x_n) - f(x^*)]$ as exact rational	Exact subtraction eliminates cancellation
Richardson extrapolation	Combine approximations at different step sizes to eliminate error terms	$h_1 = 1/3, h_2 = 1/5$: combination involves $1/(h_1^2 - h_2^2)$ with odd denominators	Exact: $[1, h_1^2 - h_2^2, 0]$ as exact rational	Exact rational coefficient
Romberg integration	Triangular array of Richardson extrapolations	Each level divides by differences of step sizes squared	Exact: all step sizes rational; all differences exact	Exact chain of extrapolations
Adams-Bashforth coefficients	Coefficients from Lagrange interpolation on $\{0, -1, -2, \dots, -(k-1)\}$	Coefficients involve $1/k!$ and products of small integers	Exact rational Lagrange coefficients	VDR-2 gym 2 (polynomial interpolation)
Condition number estimation	$\kappa(A) = \ A\ \ A^{-1}\ $ for rational A	A^{-1} computed approximately; κ is approximate	A^{-1} exact; κ exact (for exact norms)	Exact matrix inverse

Problem	Computation	Decimal failure	VDR solution	Key insight
Lebesgue constant	$\Lambda_n = \max x \Sigma$	$\ell_i(x)$	for Lagrange basis	Lagrange basis functions at rational nodes: rational function evaluation
Runge phenomenon quantification	Error growth for equidistant polynomial interpolation	Interpolation error at rational points computed exactly	Exact polynomial evaluation; exact error	Exact polynomial arithmetic
Quadrature weight verification	Gaussian quadrature weights sum to interval length	Weights involve ratios of Legendre polynomial values at irrational nodes	Weights at rational-approximation nodes: exact; sum verification exact	Exact sum check

Appendix S — Additional Domains: Coding Theory Beyond Galois Fields

Problem	Computation	Decimal failure	VDR solution	Key operation
LDPC belief propagation	Message updates as log-likelihood ratios	LLR computed from exact channel probabilities; tanh of LLR truncated	Exact channel probabilities; tanh via Taylor functional remainder	Exact probability propagation
Turbo code interleaver analysis	Permutation distances and correlation	Integer arithmetic on permutation indices	Exact integer	Trivial
Polar code construction	Bhattacharyya parameters $Z(W^{N^{(i)}})$ via recursive formula	Z values are rational for BSC; recursion $Z' = 2Z - Z^2$ truncated at each step	Exact: $[2Z - Z^2, 1, 0]$ exact rational at every recursion	Exact rational recursion

Problem	Computation	Decimal failure	VDR solution	Key operation
Convolutional code free distance	Enumerate paths through trellis; weight = sum of branch metrics	Integer branch metrics; free distance is minimum weight	Exact integer	Trivial
Algebraic geometry codes	Evaluate rational functions at rational points on curves	Evaluation involves exact rational arithmetic on curve coordinates	Exact evaluation; exact syndrome computation	Exact rational evaluation
Fountain codes (LT/Raptor)	Degree distribution $\Omega(x) = \sum \Omega_i x^i$ with rational Ω_i	Ideal soliton: $\Omega_i = 1/(i(i-1))$ has odd denominators	$[1, i(i-1), 0]$ exact; CDF sums exact	Exact probability arithmetic
Code concatenation analysis	Inner code $\times$ outer code error exponents	Products of rational error probabilities across chain	Exact rational probability chain	Exact multiplication

Appendix T — Additional Domains: Mathematical Logic and Model Theory

Problem	Computation	Decimal failure	VDR solution	Key property
Compactness theorem application	Finite satisfiability checking with rational-valued models	Model values as rationals; formula evaluation exact or not	Exact formula evaluation; satisfaction is decidable	Exact comparison
Löwenheim-Skolem cardinality	Counting models up to isomorphism; size arguments	Integer arithmetic	Exact integer	Trivial
Ultraproduct construction	Quotient by ultrafilter; equivalence of rational sequences	Pointwise rational arithmetic; equivalence requires exact comparison	Exact rational at each index; equivalence decidable	Exact comparison

Problem	Computation	Decimal failure	VDR solution	Key property
Decidability of real closed fields (Tarski)	Quantifier elimination over ordered fields	Involves sign determination of polynomials at algebraic numbers	Exact polynomial evaluation at rational approximations of algebraic numbers	Exact polynomial evaluation
Non-standard analysis arithmetic	*Q hyperrationals; transfer principle preserves exact arithmetic	Standard rationals embedded exactly; transfer preserves identities	VDR rationals satisfy transfer principle for rational identities	Exact arithmetic
Presburger arithmetic verification	Decidable theory of (Z, +, <)	Integer arithmetic with comparison	Exact integer comparison	Trivial
Peano arithmetic models	Verify models satisfy axioms	Successor, addition, multiplication on exact integers	Exact integer arithmetic	Trivial

Appendix U — Additional Domains: Actuarial Science and Insurance

Problem	Computation	Decimal failure	VDR solution	Regulatory requirement
Life table mortality rates	$q_x = d_x / l_x$; death count / survivors	$d_x = 47$, $l_x = 10000$: $q_x = 47/10000$ terminates. But $d_x = 47$, $l_x = 9953$: $q_x = 47/9953$ repeats	[47, 9953, 0] exact	Solvency II: deterministic
Net premium calculation	$P = A/\ddot{a}$ where $A = \sum v^k \cdot q_{x+k}$, $\ddot{a} = \sum v^k \cdot p_{x+k}$	$v = 1/(1+i)$; if $i = 3\%$ then $v = 100/103$ repeats (103 is prime)	[100, 103, 0] exact; chains of powers exact	GAAP: auditable

Problem	Computation	Decimal failure	VDR solution	Regulatory requirement
Reserve calculation	$V_t = A \{x+t\} - P \cdot \ddot{a} \{x+t\}$	Subtraction of two nearly-equal present values; cancellation	Exact subtraction; no cancellation	Regulatory capital
Loss development triangle	Chain-ladder factors $f_k = \Sigma c \{i, k+1\} / \Sigma c \{i, k\}$	Sums of claims; ratios with arbitrary denominators	Exact rational sums and ratios	SOX compliance
Credibility weighting	$Z = n/(n+k)$ where $k = \sigma^2/\tau^2$	If $n=7$, $k=13/3$: $Z = 7/(7+13/3) = 21/34$ repeats (17 is factor)	[21, 34, 0] exact	Actuarial standard of practice
DFA stochastic scenarios	Monte Carlo with exact rational transition probabilities	Same as Monte Carlo in VDR-27 §6.1	Exact path arithmetic; statistical error separable	Enterprise risk management
Annuity certain	$\ddot{a}_n = (1-v^n)/d$ where $d = i/(1+i)$	$d = 3/103$ for 3% interest; $v^n = (100/103)^n$; chain of powers with prime denominator	Exact rational power; exact subtraction; exact division	Policy pricing
Aggregate loss distribution	Panjer recursion: $f(x) = (1/f(0)) \Sigma (a+b \cdot k/x) \cdot p(k) \cdot f(x-k)$	Recursive formula with rational parameters; chain of multiply-and-add	Exact rational recursion; each step exact	Reserving

Appendix V — Additional Domains: Music Theory and Acoustics

Problem	Computation	Decimal failure	VDR solution	Key insight
Just intonation intervals	Perfect fifth = $3/2$; major third = $5/4$; minor seventh = $7/4$	$3/2 = 1.5$ terminates; but $7/4 \times 3/2 = 21/8 = 2.625$ terminates; chains eventually hit odd composites	Every interval exact; all comparisons exact	Exact ratio
Pythagorean comma	$(3/2)^{12} / 2^7 = 531441/524288$	$= 1.01364\dots$ terminates (all factors are 2,3) but comparing to syntonic comma $81/80$ requires exact	$[531441, 524288, 0]$ vs $[81, 80, 0]$; exact comparison	Exact comparison
Equal temperament approximation error	$2^{(7/12)}$ vs $3/2$; difference = $3/2 - 2^{(7/12)}$	$2^{(7/12)}$ is irrational; Newton approximation at depth	Newton $2^{(7/12)}$ via $\exp(7 \cdot \ln(2)/12)$ Taylor; subtract $[3,2,0]$; exact residual	Functional remainder
Harmonic series partial sums	$\sum 1/k$ for $k=1..n$	Every term with odd k in denominator repeats in decimal	Each $[1, k, 0]$ exact; sum exact	Exact rational sum
Consonance measure	Euler's gradus suavitatis: $\Gamma(n) = 1 + \sum (p_i - 1)e_i$ for $n = \prod p_i^{e_i}$	Integer factorization and arithmetic	Exact integer	Trivial
Tuning system comparison	Meantone vs Pythagorean vs just: difference in cents	Cent = $1200 \cdot \log_2(\text{ratio})$; log via Taylor on exact ratio	Exact ratio as input to Taylor log; result exact per depth	Functional remainder

Problem	Computation	Decimal failure	VDR solution	Key insight
Beat frequency		$f_1 - f_2$	where $f_1 = 440 \cdot (3/2)$, $f_2 = 440 \cdot 2^{(7/12)}$	f_1 exact rational; f_2 irrational; difference needs exact subtraction
Spectral analysis of intervals	DFT of waveform sampled at rational times	DFT roundtrip exact in VDR (VDR-13)	Frequency content exactly recoverable	VDR-13 DFT

Appendix W — Additional Domains: Economics and Econometrics

Problem	Computation	Decimal failure	VDR solution	Key operation
Input-output analysis (Leontief)	$x = (I - A)^{-1}d$; A has rational technical coefficients	Matrix inverse with denominators from all sectors' coefficients	Exact inverse; exact output vector	Exact matrix inverse
Gini coefficient	$G = (\sum_i \sum_j$	$y_i - y_j$	$)/(2n^2 \cdot \bar{y})$	Numerator is sum of absolute differences; denominator involves mean
OLS regression coefficients	$\beta = (X^T X)^{-1} X^T y$ for rational X, y	$X^T X$ inverse has denominators from all data; truncation through chain	Exact matrix product; exact inverse; exact coefficients	Exact linear algebra
Elasticity computation	$\varepsilon = (\partial Q / \partial P) \cdot (P/Q)$ $= (\Delta Q / \Delta P) \cdot (P/Q)$	Finite difference of rational observations; ratio of rationals	Exact difference; exact ratio; exact product	Exact rational chain

Problem	Computation	Decimal failure	VDR solution	Key operation
Present value of annuity	$PV = C \cdot (1 - (1+r)^{-n})/r$	If $r = 1/12$ (monthly of 100% annual): $(1+1/12)^{-n} = (12/13)^n$; 13 is prime	$[12, 13, 0]^n$ exact; entire formula exact	Exact rational power
Arrow-Debreu equilibrium	Fixed point of excess demand; involves utility maximization with rational parameters	Rational utility parameters; budget constraints involve price ratios	Exact excess demand at rational prices; exact fixed-point verification	Exact evaluation
Consumer/producer surplus	Area under/above demand/supply curves = definite integral of rational function	Exact via partial fractions (Section 17 of paper)	Exact antiderivative evaluation	Exact polynomial evaluation
Heckscher-Ohlin factor prices	Factor price equalization: solve $p = A \cdot w$ for rational A , p	Exact linear system solve	Exact solution $[w_i, 1, 0]$	Exact linear algebra

Appendix X — Additional Domains: Formal Verification

Problem	Computation	Decimal failure	VDR solution	Key property
Model checking (rational-timed)	Timed automaton with rational clock constraints	Clock comparison: is $3/7 < 5/11$? Cross-multiply: 33 vs 35; yes	Exact comparison	Exact comparison
Abstract interpretation (numerical)	Interval abstract domain: $[a, b]$ with rational bounds	Widening/narrowing on rational intervals; meet/join exact	Exact interval arithmetic	Exact comparison and arithmetic

Problem	Computation	Decimal failure	VDR solution	Key property
Loop invariant verification	Verify $i \cdot (i-1)/2 = \sum k$ for $k=0..i-1$ at each iteration	Integer arithmetic	Exact integer	Trivial
Floating-point program verification	Verify that float program satisfies spec within error bounds	Error bounds as exact rationals; float semantics modeled exactly	ULP computation as exact rational; error propagation exact	Exact rational error model
Theorem prover kernel	Type-checking and term reduction with exact coefficients	Definitional equality requires exact comparison	VDR values compared exactly; reduction produces exact results	Exact structural equality
SMT solver arithmetic theory	Quantifier-free theory of rationals: satisfiability	All operations on rationals must be exact; comparison decidable	VDR provides exact arithmetic and comparison	Exact arithmetic
Probabilistic model checking	PCTL model checking: compute reachability probability exactly	Probability as exact rational; matrix method for Markov chain	Exact probability via exact linear system solve	Exact linear algebra
Refinement checking	Implementation refines specification: simulation relation	Trace comparison at rational-valued observations	Exact comparison of observation values	Exact comparison

Appendix Y — Prime Factor Density in Mathematical Constants and Formulas

Constant/Formula	Exact rational num	Denominator	Prime factors	Decimal period	VDR repre- sentation
Euler- Mascheroni partial sum H_{10}	7381/2520	2520	$2^3 \cdot 3^2 \cdot 5 \cdot 7$	6 (from 7)	[7381, 2520, 0]
B_2 (Bernoulli)	1/6	6	2·3	1	[1, 6, 0]
B_4	-1/30	30	2·3·5	1	[-1, 30, 0]
B_6	1/42	42	2·3·7	6	[1, 42, 0]
B_8	-1/30	30	2·3·5	1	[-1, 30, 0]
B_{10}	5/66	66	2·3·11	2	[5, 66, 0]
B_{12}	-691/2730	2730	2·3·5·7·13	12	[-691, 2730, 0]
B_{14}	7/6	6	2·3	1	[7, 6, 0]
B_{16}	-3617/510	510	2·3·5·17	16	[-3617, 510, 0]
B_{18}	43867/798	798	2·3·7·19	18	[43867, 798, 0]
B_{20}	- 174611/330	330	2·3·5·11	2	[-174611, 330, 0]
RK4 coefficient 1/6	1/6	6	2·3	1	[1, 6, 0]
RK4 coefficient 1/3	1/3	3	3	1	[1, 3, 0]
Dormand- Prince 12/13	12/13	13	13	6	[12, 13, 0]
Fehlberg 4243/5348	4243/5348	5348	$2^2 \cdot 7 \cdot 191$	190	[4243, 5348, 0]
Gauss quadrature weight (n=3)	5/9	9	3^2	1	[5, 9, 0]
Gauss quadrature weight (n=3)	8/9	9	3^2	1	[8, 9, 0]
Simpson's rule 1/3	1/3	3	3	1	[1, 3, 0]
Simpson's rule 4/3	4/3	3	3	1	[4, 3, 0]

Constant/Form	Exact rational form	Denominator	Prime factors	Decimal period	VDR representation
Weddle's rule 3/10	3/10	10	2·5	0 (terminates)	[3, 10, 0]
Boole's rule 7/90	7/90	90	2·3 ² ·5	1	[7, 90, 0]
Stirling approx correction 1/12n	1/(12n)	12n	2 ² ·3·n	varies	[1, 12n, 0]
Wallis product partial (n=5)	32768/31185	31185	3·5·2079=3·5·3·693=3·5·3·3·7·11	16944/31185=0.5434090734603132768	[32768, 31185, 0]

Every Bernoulli number from B_2 onward has at least one odd prime factor in its denominator. Every RK method with order ≥ 2 has at least one coefficient with a factor of 3. Every Gaussian quadrature rule with ≥ 3 points has weights with odd prime denominators. The decimal trap is endemic in numerical mathematics.

Appendix Z — Complete Domain Coverage Summary

Category	Domains in paper	Domains in appendices	Total domains	Key shared property
Number theory	3 (continued fractions, Bernoulli, Farey)	2 (algebraic number theory, Pell equation)	5	Exact rational arithmetic on number-theoretic objects
Linear algebra	3 (Hilbert, eigenvalue, Gaussian elim)	1 (condition number)	4	Exact matrix operations; exact zero-testing
Polynomial algebra	3 (GCD, Padé, Cayley-Hamilton)	2 (Groebner, resultants)	5	Exact polynomial arithmetic; exact zero-testing
Probability/statistics	2 (Bayesian, Markov)	1 (actuarial)	3	Exact rational probability chains

Category	Domains in paper	Domains in appendices	Total domains	Key shared property
Cryptography/coding	2 (Galois field, quantum stabilizer)	2 (elliptic curve, LDPC/polar codes)	4	Exact modular/field arithmetic
Physics	1 (partition function)	2 (mathematical physics, differential geometry)	3	Exact Boltzmann weights; exact geometric quantities
Numerical methods	2 (Runge-Kutta, integration)	2 (Richardson, convergence order)	4	Exact method coefficients; exact error measurement
Algebraic geometry	0	2 (curves, Groebner application)	2	Exact zero-testing; exact point arithmetic
Topology	0	1 (homology, Smith normal form)	1	Exact integer matrix operations
Representation theory	0	1 (characters, Young tableaux)	1	Exact factorial arithmetic; exact orthogonality
Information theory	0	1 (entropy, KL, coding)	1	Exact probability $\times$ transcendental chains
Logic/verification	0	2 (model theory, formal verification)	2	Exact comparison; exact arithmetic
Economics	0	1 (Leontief, OLS, surplus)	1	Exact rational chain in economic models
Music theory	0	1 (intervals, tuning, commas)	1	Exact frequency ratios
Combinatorial optimization	0	1 (simplex, assignment, flow)	1	Exact comparison for tie-breaking

Category	Domains in paper	Domains in appendices	Total domains	Key shared property
Category theory	0	1 (functors, natural transformations)	1	Exact composition verification
Lattice algorithms	1 (LLL)	0	1	Exact threshold comparison
Total	20	23	43	Exact rational arithmetic eliminates decimal truncation

43 domains across pure mathematics, applied mathematics, computer science, physics, engineering, finance, and logic. In every domain, the structural vulnerability is the same: intermediate values with denominators containing prime factors other than 2 and 5 force decimal truncation. VDR eliminates the truncation by representing the denominator as an exact integer. The solution is not more digits. It is exact representation.

[[HOWL-VDR-28-2026](#)] Exact Rational Arithmetic for Sequential Computation. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-28-2026>

[[HOWL-VDR-1-2026](#)] VDR Arithmetic: Value, Decimal, Remainder. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-1-2026>

[[HOWL-VDR-2-2026](#)] VDR Gym: Exact Arithmetic Across Fifteen Domains. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-2-2026>

[[HOWL-MATH-3-2026](#)] The Transcendental Hierarchy. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH-3-2026>

[[HOWL-MATH-4-2026](#)] The Universal Power-of-Two Basis. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH-4-2026>

[[HOWL-VDR-14-2026](#)] VDR-LLM-Prolog. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-14-2026>

[[HOWL-VDR-21-2026](#)] VDR-LLM-Prolog on FPGA. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-21-2026>

[[HOWL-VDR-22-2026](#)] VDR-LLM-Prolog on Dedicated Silicon. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-22-2026>

[[HOWL-VDR-23-2026](#)] VDR-LLM-Prolog: Functional Remainder Hardware. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-23-2026>

[[HOWL-VDR-24-2026](#)] LLM Software. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-24-2026>

[[HOWL-VDR-25-2026](#)] LLM Server Software. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-25-2026>

[[HOWL-VDR-26-2026](#)] VDR and Diffusion. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-26-2026>

[[HOWL-VDR-27-2026](#)] VDR Beyond Language Models. Github: <https://github.com/ghowland/papers/tree/main/papers/VDR/HOWL-VDR-27-2026>